Md. Irfanur Rahman Rafio

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Research Interests

Adversarial attacks in computer vision; secondary interests in continual and zero-shot learning.

Education

Islamic University of Technology (IUT) — Gazipur, Bangladesh
BSc in Computer Science and Engineering — Jan 2020 – Jun 2024

- CGPA: **3.88/4.00**, ranked **9th** in the class.

Research Experience

A Handful of Patches: The Low-Rank, Decoupled Vulnerability of NR-IQA — Jul 2025 – Present

Work in progress, supervised by Prof. Md. Hasanul Kabir, IUT

- Constrained attack: per-image coefficients over a small global dictionary of learned 16×16 patch atoms, shared across all images and models, under an $L_\infty \leq 8/255$ budget.
- Across HyperIQA (CNN), MUSIQ (transformer) and CLIP-IQA+ (vision-language), **four learned atoms recover 63–98%** of the PGD-achievable score change in both directions, well beyond a random dictionary of the same size.
- Within the same basis, one perturbation realizes **all 27** raise, lower and hold targets across the three models, while held models drift by less than 0.01. The shared vulnerability is low-rank, and each model's response within it is decoupled from the others.
- My role: contributed ideas in group brainstorming and proposed the **tritmask experiment** behind the 27-target result, which revealed the decoupling.

A CNN-based Pipeline using Plane-wise Ensemble Technique for Classifying Alzheimer's Disease from 3D MRI Images — Jan 2023 – Jun 2024

Undergraduate thesis, IUT Computer Vision Lab; supervised by Prof. Md. Hasanul Kabir and Sabbir Ahmed

- Classified Alzheimer's disease from 3D MRI volumes with an ensemble of 2D CNNs, one per anatomical plane, using projector functions to turn each volume into 2D images.
- Reached **87.1% accuracy** on ADNI with a patient-level data split, at a much lower computational cost.

Honors and Awards

- **5th** of 309 teams, ICPC Asia Dhaka Regional Contest 2024; first to solve Problem C — 2024
- **Honorable Mention**, ICPC Asia West Continent Final Contest — 2023, 2024
- Qualified for the Rayan 2025 World Finals, Sharif University of Technology — 2024
- Codeforces Expert, maximum rating **1892**; 1200+ problems solved across online judges
- **OIC Scholarship** (partial, USD 13,500), for the top 20 IUT CSE admission test entrants — 2019

Problem Setting and Judging

Set, tested or coordinated **16+ national programming contests** and wrote **28+ original problems**, including problems for the **ICPC Asia Dhaka Regional** and its preliminary.

Selected problems

- **ICPC Asia Dhaka Regional 2025** (313 teams): **Love Marriage**, probability over a stopping process with prefix preprocessing, and **Mission Hexa**, hexagonal-lattice geometry with Euler's totient function.
- **ICPC Dhaka Regional Preliminary 2025** (1,718 teams): **Your Next Line Is, "What A Cool Problem!"**, an ad-hoc strategy game on a Hangman variant.
- **KUET IUPC 2025** (154 teams): **Perpendicular Parking**, constructive, with a proven move bound.
- **NDUB IUPC 2026** (86 teams): **86**, with Mohidul Haque Mridul; expectation via SOS DP; unsolved.
- **IUT IUPC 2026** (112 teams): **Mode Zero**, ad-hoc, on mode reductions; solved by 1 team.

Bangladesh Association of Problem Setters (BAPS) Bangladesh
Problem Setter and Judge Dec 2024 – Present

- Wrote 4 original problems and tested 14 for the ICPC Asia Dhaka Regional 2025 and its preliminary.
- Built the plagiarism checker for the ICPC Dhaka Regional 2025 Preliminary, which flagged 251 submissions from 124 teams.

Bangladesh Competitive Programming Society (BCS) Bangladesh
General Moderator Apr 2025 – Present

- Co-coordinator, problem setter and judge of IUT IUPC 2026 (112 teams); wrote 4 of its 12 problems.

Teaching Experience

Islamic University of Technology (IUT) Gazipur, Bangladesh
Part-time Lecturer, Department of CSE Sep 2024 – Present

- Teach Structured Programming II, Pattern Recognition, Algorithms and Artificial Intelligence labs.
- Redesigned Pattern Recognition Lab around how models work inside. Students build k-NN, linear and neural network classifiers on a small PyTorch-like library with a hand-written autograd Tensor, then continue in PyTorch and study a CNN with hand-assigned weights.
- Restructured the Structured Programming II Lab syllabus, wrote practice tasks for every topic and built a small presentation library on Manim to animate the lectures.
- Created a modular LaTeX template for lab assignments, now used by every programming lab in the department, and an offline local judge with test generation and stress testing, adopted by other labs.

Competitive Programming Trainer Jan 2023 – Present

- Teach weekly classes; mentored the best all-girls' team at the ICPC Asia Dhaka Regional 2024.

United International University (UIU) Dhaka, Bangladesh
Lecturer, Department of CSE Nov 2024 – Feb 2026

- Taught structured programming, OOP, data structures and algorithms, and data visualization, with labs.
- Designed the **first offering of Data Visualization** and its lab: syllabus, lectures, lab tasks and exams.
- Organized the UIU CSE Fest 2025 IUPC (150+ teams), and set problems for it and judged it through BAPS.

Industry Experience

Therap (BD) Ltd. Dhaka, Bangladesh
Associate Software Engineer (Development) Jul 2024 – Oct 2024

- Trained in object oriented design, Java, JDBC, Hibernate and JSP.

BinduLogic Limited Dhaka, Bangladesh
Software Engineer Intern Jun 2023 – Sep 2023

- Built BinduHealth OPD's backend (TypeScript, Prisma); improved BinduExam's script checker with vector geometry and graph traversal.

Selected Projects

Biscuit Rating and IUPC Slot Allocation System Live site Code Jan 2025 – Present
Rates 1,400+ contestants and 130+ institutions from 20 onsite contests in Bangladesh, and allocates contest slots from the institution ratings. **Almost every national IUPC** has used it since 2025.

Technical Skills

- **Programming Languages:** Python, C/C++, Bash, Java, TypeScript
- **Machine Learning:** PyTorch, NumPy, scikit-learn, pandas, OpenCV, Matplotlib, Jupyter
- **Tools:** Git, Linux, Docker, LaTeX, TikZ